

# HE-OFT

One-shot federated fine-tuning under multiparty CKKS

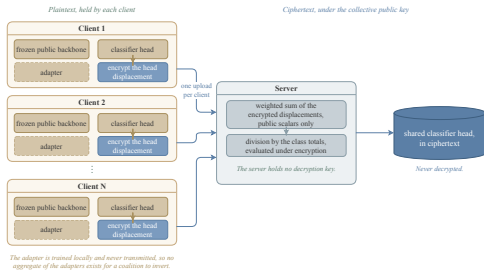
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Threat model · Paper updates · Results

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- **Clients.** They hold private data. They train locally.
- **Server.** It adds ciphertexts. It holds no key.
- **Serving party.** It answers queries. It holds no key.



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## Each client sends one message.

The message is the encrypted head displacement.

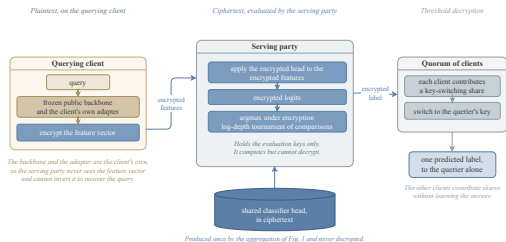
The adapter stays on the client.

- Every client message is a ciphertext.
  - The shared head stays a ciphertext. No party decrypts it.
  - The class totals stay encrypted. The server inverts them under encryption.
  - Decryption needs a share from every client.
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**Collusion.** A group of  $N - 1$  clients plus the server cannot decrypt.

- No adapter sum exists. The group has nothing to subtract.
- The head is never plaintext. The group cannot invert it.
- The totals are never plaintext. The group cannot read the last histogram.

- The client computes its own features.
- It sends them encrypted.
- The serving party applies the head.
- It takes the argmax under encryption.
- A quorum sends the label to the asker alone.



The client learns the labels it asks for.  
**This is the only channel left.**

**We return an index. We do not return scores.**

- The head is linear in features the client computes.
- Scores give a linear system.  $d + 1$  queries solve it. Here  $d + 1 = 769$ .
- An index gives about  $\log_2 C$  bits. The adversary must search.

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**Carlini et al., ICML 2024.** They stole the final projection layer of a production language model for under \$20.

- Their attack needs log-probabilities *and* a logit bias.
- OpenAI and Google then restricted that combination.
- We never offer it. This is construction, not policy.

We measured it. The adversary is a client. It queries and fits a linear model.

| Task      | $C$ | majority | $10^3$ | $5 \cdot 10^3$ | $2 \cdot 10^4$ | $5 \cdot 10^4$ | $2 \cdot 10^5$ |
|-----------|-----|----------|--------|----------------|----------------|----------------|----------------|
| AG-News   | 4   | 0.488    | 0.635  | 0.823          | 0.936          | 0.973          | 0.993          |
| DBpedia   | 14  | 0.250    | 0.387  | 0.612          | 0.804          | 0.901          | 0.971          |
| Banking77 | 77  | 0.083    | 0.174  | 0.355          | 0.592          | 0.755          | 0.900          |

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- Fidelity means agreement with the served model, not task accuracy.
  - Cost tracks the head size  $Cd$ . Fidelity 0.90 costs 3 to 5 queries per parameter.
  - A deployment sets its query allowance from  $C$  and  $d$ .
  - Small label spaces are the tight case. A small head is cheap to copy.

| Answer       | $\epsilon$ | Model accuracy | Copy fidelity |
|--------------|------------|----------------|---------------|
| index        | none       | 0.649          | 0.989         |
| index        | 2          | 0.494          | 0.889         |
| index        | 0.5        | 0.303          | 0.598         |
| distribution | none       | 0.649          | 1.000         |
| distribution | 8          | 0.266          | 0.484         |

AG-News, mean over three seeds,  $10^5$  queries.

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- The honest user pays the noise on every query.
  - The adversary averages the noise away over many queries.
  - Damage grows with the class count. The model breaks before the attack does.
  - The query allowance is the control. Noise is not a substitute.

## Threat model

- The model is never disclosed.
- Queries return one label.
- Prior work now cited for this channel.

## New measurements

- Extraction cost against query count.
- Encrypted reciprocal: 4.21 s.
- Bootstrapping keys: 15.5 MiB, once.
- Threshold decryption against  $N$ .
- Label skew and local steps sweep.

## Presentation

- Peer comparison moved to prose.
- We compare the cost of the privacy mechanism.
- We do not claim the raw accuracy margin.

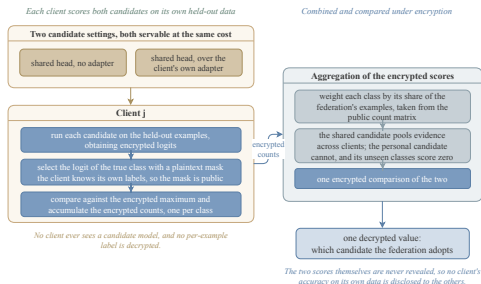
## Why

- Prior work trains a small network from scratch.
- We fine-tune a frozen public backbone.
- A table would suggest a level comparison.



| Task      | C   | servable     |                  | alone | disclosed |
|-----------|-----|--------------|------------------|-------|-----------|
|           |     | shared head  | personal adapter |       |           |
| AG-News   | 4   | <b>0.649</b> | 0.479            | 0.475 | 0.739     |
| TREC      | 6   | <b>0.607</b> | 0.429            | 0.400 | 0.711     |
| DBpedia   | 14  | <b>0.789</b> | 0.754            | 0.451 | 0.925     |
| Banking77 | 77  | 0.206        | <b>0.686</b>     | 0.249 | 0.760     |
| CIFAR-100 | 100 | 0.748        | <b>0.774</b>     | 0.197 | 0.784     |

- Never disclosing the model costs 0.03 to 0.14.
- Two arrangements are servable. Neither wins on every task.
- The federation picks one without decrypting either.
- The rule is right on 13 of 15 cells. It uses no test set.



## The direct vote fails.

- Each client scores both on its own data.
- That data follows the client's own skew.
- The vote picks the personal adapter every time.
- It is wrong on 9 of 15 cells.

## The estimator works.

- Score both under the federation class prior.
- The shared head pools evidence. The personal one cannot.
- Unseen classes score zero. This is a lower bound.
- One value is decrypted: the winner index.

## Per operation

|                         |         |
|-------------------------|---------|
| head on encrypted query | 35.2 ms |
| head on plaintext query | 4.8 ms  |
| rotation                | 30.8 ms |
| encrypted reciprocal    | 4.21 s  |
| key switch, $N = 10$    | 181 ms  |

We implement the protocol in real multiparty CKKS. We do not simulate it.

## Communication, $N = 10$

|                      |                |
|----------------------|----------------|
| setup keys, once     | 91 MiB         |
| bootstrap keys, once | 15.5 MiB       |
| per query, total     | <b>5.0 MiB</b> |

- Threshold decryption is linear in  $N$ : 18 ms and 2.0 MiB per client.
- Local refresh replaces 510 MiB per query with 15.5 MiB once.
- A GPU gives about  $5\times$  on the primitives at matched ring degree.

| Venue                | Next deadline | Link                                                                    |
|----------------------|---------------|-------------------------------------------------------------------------|
| NDSS 2027            | 19 Aug 2026   | <a href="https://ndss-symposium.org">ndss-symposium.org</a>             |
| USENIX Security 2027 | 25 Aug 2026   | <a href="https://usenix.org">usenix.org</a>                             |
| IEEE S&P 2027        | 17 Nov 2026   | <a href="https://sp2027.ieee-security.org">sp2027.ieee-security.org</a> |
| ACM CCS 2027         | not announced | <a href="https://sigsac.org">sigsac.org</a>                             |
| AsiaCCS 2027         | 21 Aug 2026   | <a href="https://sec-deadlines.github.io">sec-deadlines.github.io</a>   |
| PETS 2027            | 31 Aug 2026   | <a href="https://petsymposium.org">petsymposium.org</a>                 |
| DSN 2027             | 2 Dec 2026    | <a href="https://sec-deadlines.github.io">sec-deadlines.github.io</a>   |
| EuroS&P 2027         | see call      | <a href="https://ieee-security.org">ieee-security.org</a>               |
| IEEE TIFS            | rolling       | <a href="https://ieee.org">ieee.org</a>                                 |
| IEEE TNSE            | rolling       | <a href="https://ieee.org">ieee.org</a>                                 |

Dates from [sec-deadlines.github.io](https://sec-deadlines.github.io), checked 29 July 2026. Confirm on the venue page.

## Open points

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- Client count sweep is still running.
- Pooled ceiling gives the reference the tables need.
- Matched partitions for the private baseline are queued.